

7 When an impulse arrives at a neuromuscular junction, it stimulates a muscle fibre of striated muscle to contract.

(a) (i) Outline the **similarities** in structure between a neuromuscular junction and a cholinergic synapse.

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..... [4]

(ii) The hydrolysis of ATP during muscle contraction releases inorganic phosphate (P_i).

Calcium ions (Ca^{2+}) can combine with P_i in the sarcoplasmic reticulum to form insoluble calcium phosphate. This may result in fewer power strokes occurring in sarcomeres.

Suggest why calcium phosphate formation in the sarcoplasmic reticulum may result in fewer power strokes occurring in sarcomeres.

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- (b) Adrenaline is a hormone that can affect muscle contraction. Adrenaline binds to G-protein-coupled receptors on T-tubule membranes.

The cell signalling pathway that occurs in response to the binding of adrenaline is similar to the pathway that occurs in liver cells in response to the binding of glucagon.

Fig. 7.1 is an outline of the cell signalling pathway of adrenaline.

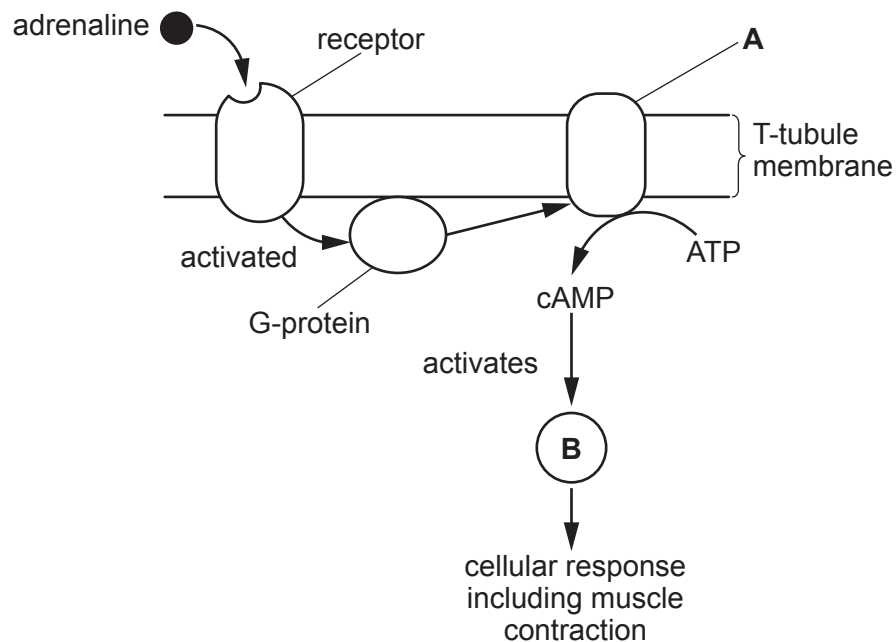


Fig. 7.1

Identify the molecules represented by **A** and **B** in Fig. 7.1.

A

B

[2]

[Total: 9]